Linearity:

Ex

$V_0 = 1V = ?$ (Find i_0)

$i_0 = \frac{1}{2} = 0.5A$, $i_0 = \frac{V_1 - 1}{4} = 0.5$

$V_1 = 3V$

$i_1 = \frac{V_1}{3} = \frac{3}{3} = 1A$

KCL @ V_1

$i_2 = 0.5A + 1A = 1.5A$

$= V_2 - V_1$

$1.5 = V_2 - 3$, $V_2 = 4.5$

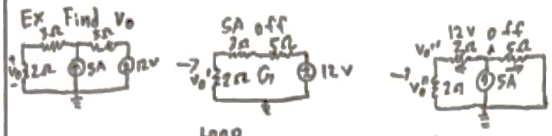
when $V_0 = 1V$, $V_S = 4.5V$

$V_S = 4V$, $\boxed{V_0 = \frac{4}{4.5} V}$

Superposition:

- Turn off all sources except 1
- Analyse as sep. circuits
- Add values

Ex Find V_0



$V_0 = V_0' + V_0''$

$= 2.4V + 6V$

$= \boxed{7.4V}$

Loop

$-12 + 10 \cdot I_1 = 0$

$10 I_1 = 12$

$I_1 = 1.2A$

$V_0' = 2.4V$

KCL @ A

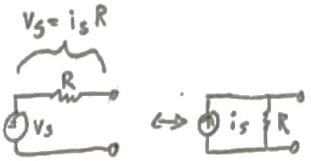
$5 = \frac{V_A}{5} + \frac{V_A}{5}$

$V_A = 12.5V$

$V_0'' = 12.5 \cdot \frac{2}{2+5} = 5V$

Source Transform:

$V_S = i_S R$

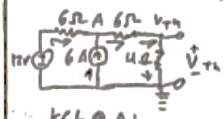


Thevenin Theorem:

$R_{Th} = \frac{V_{Th}}{I_{sc}}$

- Open circuit to Find V_{Th}
- R_{Th} from turning all independent sources off and finding $R_{Eq} \Rightarrow R_{Th}$

Ex Find Thev. equiv. + I_{sc} - I_{sc} use formula



$R_{Th} = (6+6) // 4$

$= \boxed{3\Omega}$

$I_S = \boxed{4A}$

KCL @ A:

$\frac{12V - V_A}{6} + 6 = \frac{V_A}{10}$

$60 - 5V_A + 180 = 3V_A$, $240 = 8V_A$

$V_A = 30V$

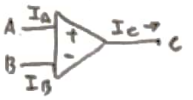
$V_{Th} = 30 \cdot \frac{4}{10} \rightarrow$ v div

$= \boxed{12V}$

Maximum Power Transfer:

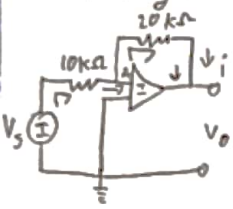
$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{I_n^2 R_{Th}}{4}$

Ideal op-amps



- $V_A = V_B$
- $I_A = I_B = 0A$
- $I_A + I_B \neq I_C$
- Always use KCL in input terminals

Ex Find gain V_0/V_S and i when $V_S = 2$



$V_A = V_B = 0$

KCL @ A

$\frac{V_S - V_A}{10k\Omega} = 0 + \frac{V_A - V_0}{20k\Omega}$

$2V_S - 2V_A = V_A - V_0$

$2V_S = -V_0$

$\boxed{\frac{V_0}{V_S} = -2}$

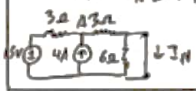
$i = \frac{V_A - V_0}{20k\Omega}$

$= \frac{0 - (-4)}{20} = \boxed{\frac{4}{20} mA}$

when $V_B = 2$, $V_0 = -4$

Norton Theorem:

Ex Find I_N & R_N



USE KCL for I_n and $R_N = R_{Eq}$

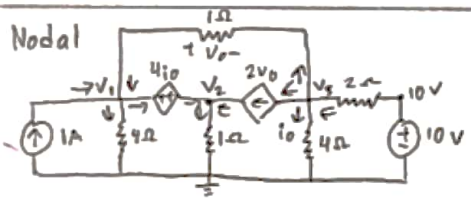
$P = V^2/R = VI = I^2 R$

$I = V/R = \sqrt{P/R} = P/V$

$R = P/I^2 = V/I = V^2/P$

$V = IR = \sqrt{PR} = P/I$

Nodal



$V_2 - V_1 = 4i_0$ $\Rightarrow V_2 - V_1 = V_3$, $V_2 = V_3 + V_1$

$V_3 = 4i_0$, $i_0 = \frac{V_3}{4}$

$V_1 - V_3 = V_0$

KCL @ V_1

$V_3 - V_1 + 1 = \frac{V_1}{4} + I_{12}$

$V_3 - V_1 + 1 = \frac{V_1}{4} + 3V_3 - V_1$

KCL @ V_2

$I_{12} + I_{32} = V_2$

$I_{12} + 2V_0 = V_2$

$I_{12} + 2(V_1 - V_3) = V_1 + V_3$

$I_{12} = 3V_3 - V_1$

KCL @ V_3

$\frac{10 - V_2}{2} = \frac{V_3}{4} + \frac{V_3 - V_1}{1} + 2V_0$

$\frac{10 - V_3}{2} = \frac{V_3}{4} + V_3 - V_1 + 2(V_1 - V_3)$

Capacitors & Inductors

Cur thru cap: $i = C\frac{dv}{dt}$ $\quad q = Cv$

volt thru ind: $v = L\frac{di}{dt}$

volt thru cap $V(t)$: $\frac{1}{C_{eq}}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$

cur thru ind: $\frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0)$

Energy stored

cap: $w = \underbrace{\frac{1}{2}CV^2}_{main} = \frac{q^2}{2C}$ $\quad$ ind: $w = \underbrace{\frac{1}{2}Li^2}_{main}$

- Cap in par: $C_{eq} = C_1 + C_2 + C_3 ... C_N$ $\quad$ ser: $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + ... \frac{1}{C_N}$

ind in par: $\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + ... \frac{1}{L_N}$

ser: $L_1 + L_2 + ... L_N$

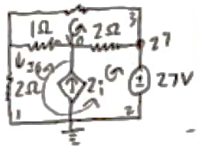
*DC = Steady-state

↳ cap is open

↳ ind is short

Supermesh

Ex



Supermesh

$2I_1 - 27 + 2(I_2 - I_3) + (I_1 - I_3) = 0$

Loop 3

$3I_3 + (I_3 - I_1) + 2(I_3 - I_2) = 0$

CS

$I_1 - I_2 = 2i_0$, $I_1 = I_0$

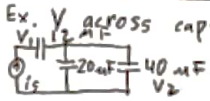
$I_1 - I_2 = 2I_1$

$I_1 + I_2 = 0$

USE Matrix to solve for unknown

$I_3 - I_2 = \frac{V_0 - 27}{2}$

Ex. V across capacitor



$i_s = 50e^{-2t}$ mA

$V_1(0) = 50V$

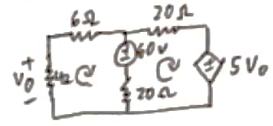
$V_2(0) = 20V$

$C_{eq} = \frac{12 \cdot 60}{12+60} = 10\mu F$

$12\mu F$: $V(t) = \left(\frac{1}{12 \cdot 10^{-6}}\int_0^t 50e^{-2t} \cdot 10^{-3}\,dt\right) + 50 = -2083e^{-2t} + 2133.33V$

$40\mu F$: $V(t) = \left(\frac{1}{60 \cdot 10^{-6}}\int_0^t 50e^{-2t} \cdot 10^{-3}\,dt\right) + 20 = -416.67$

Mesh



$V_0 = I_1 \cdot 4\Omega$

Loop 1

$60 + 20(I_1 - I_2) + 4I_1 + 6I_1 = 0$

Loop 2

$20I_2 + 5V_0 - 60 + 20(I_2 - I_1)$

Solving RC cct

1. Find $V_c(0)$, $V_c(\infty)$, $\tau = RC$
2. Use $V_c(t) = V_c(\infty) + [V_c(0) + V_c(\infty)]e^{-t/RC}$

Solving RL cct

1. Find $i(0)$, $i(\infty)$, $\tau = L/R$
2. Use $i(t) = i_{ss} + [I_0 - i_{ss}]e^{-t/\tau}$

Power dis: $p(t) = \frac{V_0^2}{R}e^{-2t/\tau}$ RC

Energy abs: $w_R(t) = \frac{1}{2}CV_0^2(1 - e^{-2t/\tau})$

RL $p(t) = I_0^2 R e^{-2t/\tau}$

$w_R(t) = \frac{1}{2}LI_0^2(1 - e^{-2t/\tau})$

CamScanner

Time domain : $v(t) = V_0 \cos(\omega t + \theta)$

Phasor : $v = V_0 \angle \theta = V_0 e^{j\theta}$

Impedence : $Z = \frac{V}{I} = R(\omega) + jX(\omega)$

Admittance : $Y = \frac{1}{Z} = G(\omega) + jB(\omega)$

↳ $Z = R$, $Z = j\omega L \rightarrow$ inductor

↳ $Z = 1/j\omega C \rightarrow$ capacitor

$Z_{eq} = Z_1 + Z_2 + Z_3, ,,$

$I = V/Z_{eq}$ ↳ series

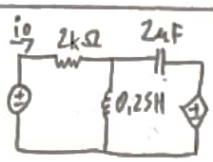
$I = I_1 + I_2 + I_3, ,,$

↳ parallel

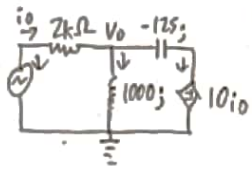
$\frac{1}{Z_{eq}} = \frac{1}{V} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} + , ,,$

$Y_{eq} = Y_1 + Y_2 + Y_3 + , ,,$

Nodal, find i_o



$\Rightarrow 25\angle 0°$



$$\frac{V_o}{1000j} + \frac{V_o - 10i_o}{-125j} + \frac{V_o - 25}{2000} = 0 \quad , \quad i_o = \frac{(25 - V_o)}{1000j}$$

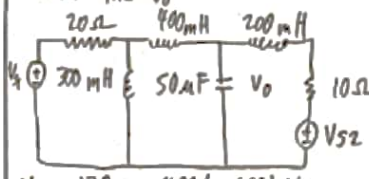
$V_o = 1,7768\angle -81,363°$

$i_o = 12,398\angle 4,06$

$= \boxed{12,398 \cos(4.10^{-3} t + 4,06°)\ mA}$

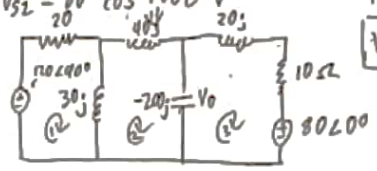
+ Flip j, conj

Mesh, find V_o



$V_{S1} = 120 \cos(100t + 90°)$ V

$V_{S2} = 80 \cos 100t$ V



Mesh 1

$-120\angle 90° + 20I_1 + 30j(I_1 - I_2) = 0$

Mesh 2

$40j I_2 - 200j(I_2 - I_3) + 30j(I_2 - I_1)$

Mesh 3

$-200j(I_3 - I_2) + 20j I_3 + 10 I_3 - 80\angle 0° I_3$

Put into matrix

$\boxed{V_o = -200j(I_2 - I_3)}$

Inst, $P = Vi$

$P_{avg} = \frac{1}{T}\int_0^T P\, dt$

If $v(t) = V_m \cos(\omega t + \theta_v)$ & $i(t) = I_m \cos(\omega t + \theta_i)$

then $V_{rms} = V_m/\sqrt{2}$ & $I_{rms} = I_m/\sqrt{2}$ and $\rightarrow$

$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{rms} I_{rms} \cos(\theta_v - \theta_i)$

Ind & Cap absorb no P. Res absorb $\frac{1}{2} I_m^2 R = I_{rms}^2 R$

Pow fac, $Pf = \cos(\theta_v - \theta_i)$

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